Supplementary Material

LOCUS: Verifying OCR by Local Optimality over Counterfactual String Graphs

Shikhar Dubey[1], **Sanhita Saxena**[2]

[1]Intelligent Systems, School of Computing and Electrical Engineering, and [2]School of Mathematical and Statistical Sciences, Indian Institute of Technology Mandi, Himachal Pradesh, India
s24130@students.iitmandi.ac.in, s25008@students.iitmandi.ac.in

This document supplements the article *LOCUS: Verifying OCR by Local Optimality over Counterfactual String Graphs*. It gives the implementation details needed for reproduction and the complete experimental record behind the summaries of the main article; references without the prefix S (e.g. "Table 3") point to the main article, and numbered citations such as [6] refer to its reference list. All tables and figures are generated by scripts from the raw result files; no number was entered by hand.

S1. Benchmark construction

Crops. FUNSD words are cropped from the page with a 2-pixel margin after rescaling the LayoutLM-normalised boxes ($[0, 1000]^2$) of the public Hugging Face release to pixel coordinates; cropping with the raw values yields misaligned crops for almost every word (the Tesseract word error rate is 99.8% without the rescaling and 36.9% with it). CORD words are cropped from the axis-aligned bounding box of each word quadrilateral; the official train and validation receipts form the training pool. SROIE, IAM and IIIT5K are distributed as crops. Crops narrower than 6 or shorter than 8 pixels, and empty transcriptions, are discarded.

Engines. Tesseract 5 (LSTM engine, `-psm 7`, crops shorter than 32 px are bicubically upscaled to 32 px) outputs the space-joined words it detects and their mean confidence. TrOCR-small (`microsoft/trocr-small-printed` or `-handwritten`) decodes greedily; its confidence is the mean token probability.

Labels. A claim is CORRECT iff it equals the ground truth after whitespace normalisation (case-insensitive for SROIE, whose ground truth is upper-case; case-insensitive and restricted to alphanumeric characters for IIIT5K, following its standard protocol). No other normalisation is applied, so case and punctuation errors count as errors, as they do for any downstream consumer that copies the field verbatim.

Caveats. TrOCR's printed checkpoints were fine-tuned on SROIE receipts; the verifier V_1 is therefore in-domain on SROIE, and TrOCR-small makes only 3–6% errors there, which is why

SROIE uses Tesseract claims. Tesseract is degenerate on handwriting (93.9% word error on IAM), which is why IAM uses TrOCR-small claims.

S2. Counterfactual neighbourhood generation

Confusion groups. Substitutions are drawn from the groups `OOoQDCc`, `Il1|!ij`, `5Ss$`, `8B&3`, `2Zz`, `6Gb`, `9gq`, `7T/`, `uvnUV`, `wW`, `.,:;’‘"`, `-_~=`, `(){}[]`, `aeo`, `hbk`, `rnm`, `cC(e`, `xX%`, `yY`, `pP`, `fft`; at most two alternatives per character are used.

Algorithm. Given claim h and verifier readings g_1 (TrOCR) and g_2 (PARSeq):

1. add g_2 and g_1 (if different from h);
2. for each reading, add every single edit of h suggested by its Levenshtein alignment to the reading: substitution of an aligned mismatching character, deletion of an unaligned character of h, insertion of an unaligned character of the reading;
3. for each position of h (up to 32): a case flip, up to two confusion substitutions, and a deletion;
4. deduplicate, keep insertion order, and stop at $M = 28$ neighbours.

No step uses the ground truth. The share of real errors whose truth is a node of the graph is reported in Table S33.

Rendering. $R(v)$ draws v in Liberation Sans (metrically equivalent to Arial) at $0.75\times$ the crop height (clamped to $[32,96]$ px), black on white, with a quarter-height margin. The same rendering is fed to both verifiers. PARSeq scores the string restricted to its 94-character charset and truncated to its maximum label length of 25.

S3. Label-free pretraining data

From the unlabeled training pool we keep images on which two independent engines produce the identical string (Tesseract and TrOCR-small; TrOCR-small and TrOCR-base on IAM), shuffle them with a fixed seed and keep at most 3,000. Each yields a pseudo-correct graph (the agreed string) and, if the corruption changes the string, a pseudo-incorrect graph built around one or two random OCR-style edits: confusable substitution (45%), case flip (15%), deletion (20%) or insertion of a confusable character (20%). The repair target of a corrupted graph is the node holding the agreed string (when present). Agreement is a proxy, not ground truth; its precision on the training pool is reported in Table S1.

Table S1. Label-free pretraining sets. Precision of engine agreement is measured post hoc against the ground truth of the training pool (never used for training).

Dataset	agreeing images	graphs	precision of agreement	truth in graph (corruptions)
FUNSD	2,561	5,085	94.5%	99.6%
CORD	3,000	5,959	98.7%	99.8%
SROIE	2,663	5,301	99.0%	100.0%
IAM	3,000	5,950	83.7%	100.0%
IIIT5K	860	1,711	99.3%	99.5%

Table S2. Hyperparameters (identical for all datasets and all k).

Node features	2×11 evidence + 7 claim deltas (tanh-squashed) + 8 edit-kind + 1 position + 6 agreement (claim node)
Edge features	8 edit kinds + sibling + path + self (one-hot)
GNN	3 GATv2 layers, hidden 64, 4 heads, residual + LayerNorm, dropout 0.2
Readout	claim state, attention pool, max pool over rivals, $\beta \log P_{\text{opt}}$ (β learned)
Pretraining	15 epochs, AdamW lr 10^{-3}, wd 10^{-4}, batch 64, $\lambda = 0.5$
Few-shot fine-tuning	300 steps of the whole pretrained GNN, AdamW lr $3 \cdot 10^{-4}$, batch = half the support set, class-balanced BCE
Inference	average of the five networks fine-tuned on the five support sets (seeds 0–4); threshold 0.5
Ablation-only heads	CV-selected prior/anchored head and linear probe (Table 4 only; not part of LOCUS)
Full supervision	60 epochs, early stopping (patience 8) on a 15% split of the training pool
Seeds	0–4 (support sampling, initialisation, pretraining)

S4. Hyperparameters and baseline protocols

Thresholds. Engine confidence and the ρ-margin rule are thresholded at the value maximising macro-F1 on the k-shot support (seed-specific). Agreement baselines are binary (accept iff equal). The graded PARSeq score is $+c$ if PARSeq's reading equals the claim and $-c$ otherwise, where c is PARSeq's sequence confidence (product of character probabilities); it is thresholded at 0, which coincides with the binary decision.

VLM judge prompt. *Does the image show exactly the text "{claim}" (same characters, case and punctuation)? Answer yes or no.* The image is passed at its native resolution (upscaled to a height of 56 px if smaller; Qwen2.5-VL pixel budget 32–256 visual tokens). The score is the renormalised probability of `yes`/`Yes` versus `no`/`No` as the first generated token.

ConfidNet-style learned confidence. A one-hidden-layer MLP (32 units, ℓ_2 10^{-2}) on the engine confidence and five likelihood statistics of the claim under TrOCR-base, trained on the support set.

S5. Additional analyses and qualitative examples

Error types

Table S3 breaks down the detection rate (recall of truly incorrect claims) by the edit that separates the claim from the truth. Operating points: LOCUS at $P = 0.5$, PARSeq-graded at its natural threshold, the VLM at 0.5, and engine confidence at its *test-optimal* threshold (favourable to the baseline).

Table S3. Detection rate (%) by error type; number of errors of that type in parentheses ($k = 25$).

Dataset	Method	substitution	insertion	deletion	case only	multi-edit
FUNSD	LOCUS	87.7 (236)	79.7 (390)	71.8 (78)	77.2 (57)	97.1 (713)
	PARSeq re-OCR (graded)	86.0 (236)	80.3 (390)	70.5 (78)	78.9 (57)	98.2 (713)
	TrOCR-base re-OCR	93.2 (236)	85.1 (390)	82.1 (78)	96.5 (57)	98.5 (713)
	Qwen2.5-VL-7B judge	62.3 (236)	37.7 (390)	67.9 (78)	26.3 (57)	79.2 (713)
	Engine confidence	74.2 (236)	76.4 (390)	48.7 (78)	71.9 (57)	89.3 (713)
CORD	LOCUS	95.3 (342)	93.2 (73)	92.5 (40)	80.0 (20)	99.1 (466)
	PARSeq re-OCR (graded)	95.6 (342)	98.6 (73)	95.0 (40)	100.0 (20)	100.0 (466)
	TrOCR-base re-OCR	98.0 (342)	100.0 (73)	95.0 (40)	90.0 (20)	100.0 (466)
	Qwen2.5-VL-7B judge	74.3 (342)	20.5 (73)	87.5 (40)	10.0 (20)	94.2 (466)
	Engine confidence	64.6 (342)	53.4 (73)	92.5 (40)	55.0 (20)	92.9 (466)
SROIE	LOCUS	97.5 (641)	76.6 (137)	72.4 (105)	–	99.3 (997)
	PARSeq re-OCR (graded)	98.9 (641)	100.0 (137)	79.0 (105)	–	99.8 (997)
	TrOCR-base re-OCR	99.5 (641)	82.5 (137)	79.0 (105)	–	99.4 (997)
	Qwen2.5-VL-7B judge	68.2 (641)	29.2 (137)	37.1 (105)	–	90.4 (997)
	Engine confidence	75.5 (641)	75.9 (137)	44.8 (105)	–	92.8 (997)
IAM	LOCUS	93.5 (505)	76.8 (95)	57.7 (26)	43.2 (185)	92.8 (1046)
	PARSeq re-OCR (graded)	95.6 (505)	98.9 (95)	57.7 (26)	94.6 (185)	98.1 (1046)
	TrOCR-base re-OCR	88.9 (505)	85.3 (95)	76.9 (26)	29.2 (185)	95.9 (1046)
	Qwen2.5-VL-7B judge	80.0 (505)	61.1 (95)	61.5 (26)	57.8 (185)	86.3 (1046)
	Engine confidence	82.8 (505)	68.4 (95)	69.2 (26)	27.0 (185)	89.5 (1046)
IIIT5K	LOCUS	92.4 (172)	96.4 (28)	95.1 (61)	–	99.8 (608)
	PARSeq re-OCR (graded)	94.2 (172)	100.0 (28)	95.1 (61)	–	99.8 (608)
	TrOCR-base re-OCR	93.6 (172)	100.0 (28)	93.4 (61)	–	99.8 (608)
	Qwen2.5-VL-7B judge	89.0 (172)	96.4 (28)	93.4 (61)	–	99.7 (608)
	Engine confidence	50.6 (172)	60.7 (28)	65.6 (61)	–	94.1 (608)

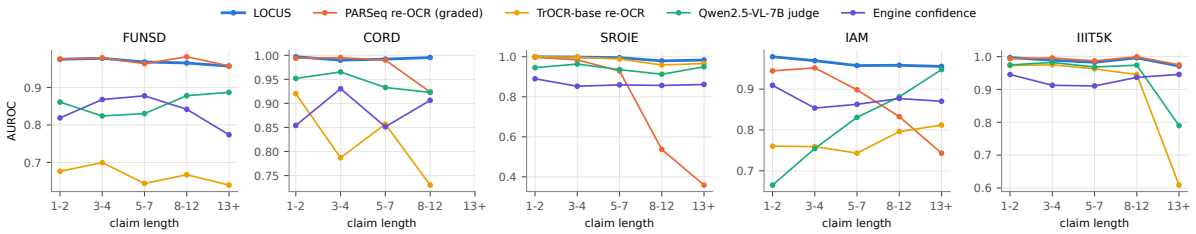


Figure S1. AUROC by claim length ($k = 25$). Bins with fewer than 30 words or a single class are omitted.

Claim length

Macro-F1 by claim length (Table S4). Accuracy decreases with claim length on every dataset, most on SROIE lines and IAM words, where long claims also hit the cap of $M = 28$ rivals (Section 5.6).

Train/test duplicates. A pixel-level hash comparison of all training and test crops finds no shared image on FUNSD, SROIE, IAM and IIIT5K. On CORD, 79 test crops are pixel-identical to training crops because some receipts occur in both official splits; removing them leaves LOCUS' macro-F1 (.978) and AUROC (.993) unchanged.

Table S4. LOCUS macro-F1 by claim length in characters ($k = 25$; number of test words in parentheses; bins with fewer than 30 words or one class omitted).

Dataset	1–3	4–7	8–15	≥16
FUNSD	0.951 (1358)	0.935 (1731)	0.940 (903)	–
CORD	0.984 (746)	0.975 (1367)	0.975 (237)	–
SROIE	0.992 (612)	0.983 (1331)	0.963 (975)	0.959 (1082)
IAM	0.933 (1789)	0.881 (1599)	0.869 (595)	–
IIIT5K	0.979 (1117)	0.973 (1453)	0.971 (404)	–

Table S5. Seed-level comparison: each of the five $k = 25$ models (support seeds 0–4) separately against the strongest baseline; difference in AUROC and in macro-F1.

		ΔAUROC, seeds 0–4					Δmacro-F1, seeds 0–4				
Dataset	baseline	0	1	2	3	4	0	1	2	3	4
FUNSD	PARSeq (graded)	-0.006	-0.012	-0.009	-0.015	-0.008	-0.028	-0.012	-0.018	-0.026	-0.021
CORD	PARSeq (graded)	+0.006	+0.007	+0.007	-0.006	+0.007	-0.019	+0.006	+0.007	-0.044	+0.008
SROIE	TrOCR-base	+0.011	+0.014	+0.011	+0.011	+0.013	-0.023	-0.011	-0.028	-0.016	-0.026
IAM	PARSeq (graded)	+0.031	+0.046	+0.042	+0.044	+0.030	+0.000	+0.015	-0.001	-0.003	-0.016
IIIT5K	PARSeq (graded)	-0.005	-0.008	-0.006	-0.010	-0.006	-0.018	-0.059	-0.024	-0.041	-0.023

Table S6. Expected calibration error (15 bins) of the correctness probability ($k = 25$; PARSeq-graded mapped to $[0,1]$ as $(s+1)/2$). Lower is better.

Dataset	LOCUS	PARSeq graded	Qwen2.5-VL-7B	Engine conf.
FUNSD	0.024	0.039	0.019	0.136
CORD	0.025	0.018	0.045	0.114
SROIE	0.020	0.197	0.057	0.213
IAM	0.030	0.058	0.110	0.289
IIIT5K	0.027	0.009	0.136	0.042

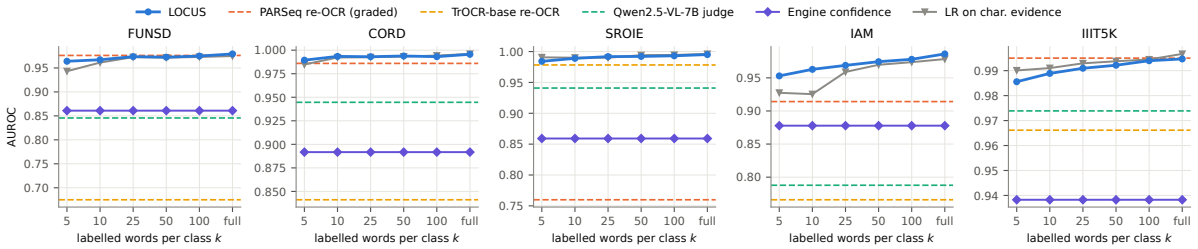


Figure S2. AUROC as a function of the number of labelled words per class (ensembles of five seeds; dashed: label-free baselines).

Table S7. Cross-dataset transfer, AUROC (rows: training dataset, fully supervised; columns: test dataset; diagonal: in-domain).

Train \ Test	FUNSD	CORD	SROIE	IAM	IIIT5K
FUNSD	.979	.995	.979	.968	.993
CORD	.968	.996	.987	.954	.986
SROIE	.941	.992	.996	.892	.992
IAM	.962	.989	.952	.985	.991
IIIT5K	.961	.993	.975	.963	.995

Calibration

AUROC and AURC learning curves

Cross-dataset transfer (AUROC)

Ablations: AUROC and repair

Table S8. Ablations at $k = 25$, AUROC (ensemble of five seeds). Last column: mean change w.r.t. the full model.

Variant	FUNSD	CORD	SROIE	IAM	IIIT5K	Δ
LOCUS (full)	.973	.993	**.992**	.968	.991	
+ CV-selected prior/anchored head	.974	.993	.990	.968	.992	-0.000
+ linear probe on embedding	.974	.993	**.992**	.969	**.993**	+0.001
w/o verifier-agreement node features	.974	.993	.991	.968	.987	-0.001
w/o edit-graph GNN (char evidence LR)	.973	.992	.990	.959	**.993**	-0.002
w/o label-free pretraining	.969	.992	.990	**.970**	.990	-0.002
w/o local-optimality term	.972	.993	**.992**	.966	.990	-0.001
w/o repair loss	.971	.993	**.992**	.965	.990	-0.001
w/o render-normalisation	.973	.993	**.992**	.967	.992	-0.000
w/o latent render distance	.973	.993	**.992**	.969	.990	-0.000
likelihood only (no render)	.974	.992	.991	.969	.992	+0.000
TrOCR verifier only	.964	.991	**.992**	.956	.991	-0.005
PARSeq verifier only	.974	.993	.990	.965	.992	-0.001
star edges only	.971	.993	**.992**	.969	.991	-0.001
no message passing	**.975**	**.994**	**.992**	.967	.991	+0.000
GCN backbone	.970	.993	.991	.966	.990	-0.002
GraphSAGE backbone	.973	.993	**.992**	**.970**	.990	-0.000
GIN backbone	.973	.991	.989	.957	.990	-0.004

Sensitivity

Table S9. Sensitivity of LOCUS ($k = 25$, ensemble of five seeds) to the size of the label-free pretraining set, the number of fine-tuning steps, and the support-selection strategy (label-free k-means clusters the unlabeled pool in the pretrained embedding space and labels the $2k$ medoids).

		FUNSD		CORD		IAM	
Factor	Value	F1	AUROC	F1	AUROC	F1	AUROC
pretrain fraction	0.1	.943	.967	.977	.993	.902	.968
pretrain fraction	0.25	**.944**	.968	.978	**.994**	**.908**	**.970**
pretrain fraction	0.5	.941	.967	.979	**.994**	.904	**.970**
pretrain fraction	1.0	.940	**.974**	**.980**	.993	.900	.967
fine-tuning steps	100	**.942**	**.976**	.975	**.993**	.884	.960
fine-tuning steps	300	.940	.974	**.980**	**.993**	**.900**	**.967**
fine-tuning steps	1000	.938	.974	.979	**.993**	.899	**.967**
support selection	random	**.940**	**.974**	**.980**	.993	.900	**.967**
support selection	label-free k-means	.935	.964	**.980**	**.994**	**.907**	.964

Main comparison at other label budgets and metrics

Composition of real OCR errors

S6. Full results

Tables S18–S32 report every method at every k with seed standard deviations, and every metric at $k = 25$.

Table S10. Macro-F1 at $k = 10$ (protocol of Table 2).

Method	FUNSD	CORD	SROIE	IAM	IIIT5K	Avg.
Baselines						
Engine MSP confidence	.722	.775	.749	.782	.817	.769
ConfidNet-style learned conf. [7]	.702	.813	.785	.783	.915	.800
Two-engine voting	.609	.772	.976	.759	.862	.796
Re-OCR: TrOCR-base [4]	.605	.811	**.979**	.758	.952	.821
Re-OCR: Florence-2 [5]	.647	.615	.478	.369	.823	.586
Re-OCR: PARSeq [3]	**.946**	.973	.742	.879	**.993**	.907
Re-OCR: PARSeq, graded	**.946**	.973	.742	.879	**.993**	.907
VLM judge: Qwen2.5-VL-7B [6]	.760	.891	.866	.700	.888	.821
VLM judge, oracle threshold†	.762	.902	.883	.719	.928	.839
Best baseline per dataset, chosen in hindsight‡	.946	.973	.979	.879	.993	.954
Strong learned baselines on our evidence						
LR on flattened graph evidence	.931	.957	.939	.881	.974	.937
LR on character evidence	.915	.970	.949	.839	.965	.927
GBDT on all evidence	.885	.960	.967	.882	.947	.928
DeepSets over rivals (no edges)	.931	.978	.930	**.896**	.964	.940
LOCUS-GNN from scratch	.920	.969	.913	.877	.961	.928
Ours						
LOCUS zero-shot (no labels)	.918	.904	.798	.825	.968	.883
LOCUS (k-shot)	.936	**.980**	.954	.888	.975	**.947**

Table S11. Macro-F1 at $k = 100$.

Method	FUNSD	CORD	SROIE	IAM	IIIT5K	Avg.
Baselines						
Engine MSP confidence	.771	.799	.777	.801	.790	.788
ConfidNet-style learned conf. [7]	.830	.905	.873	.832	.924	.873
Two-engine voting	.609	.772	.976	.759	.862	.796
Re-OCR: TrOCR-base [4]	.605	.811	**.979**	.758	.952	.821
Re-OCR: Florence-2 [5]	.647	.615	.478	.369	.823	.586
Re-OCR: PARSeq [3]	**.946**	.973	.742	.879	**.993**	.907
Re-OCR: PARSeq, graded	**.946**	.973	.742	.879	**.993**	.907
VLM judge: Qwen2.5-VL-7B [6]	.760	.891	.866	.700	.888	.821
VLM judge, oracle threshold†	.762	.902	.883	.719	.928	.839
Best baseline per dataset, chosen in hindsight‡	.946	.973	.979	.879	.993	.954
Strong learned baselines on our evidence						
LR on flattened graph evidence	.945	**.980**	.978	.918	.970	.958
LR on character evidence	.940	.979	.977	.920	.967	.956
GBDT on all evidence	.944	**.980**	**.979**	.923	.983	**.962**
DeepSets over rivals (no edges)	.942	**.980**	.976	.926	.968	.958
LOCUS-GNN from scratch	.938	.978	.977	**.927**	.973	.959
Ours						
LOCUS zero-shot (no labels)	.918	.904	.798	.825	.968	.883
LOCUS (k-shot)	**.946**	**.980**	.974	**.927**	.973	.960

Table S12. Macro-F1 with full supervision (85% of the training pool; 15% for early stopping).

Method	FUNSD	CORD	SROIE	IAM	IIIT5K	Avg.
Baselines						
Engine MSP confidence	.787	.805	.790	.807	.854	.809
ConfidNet-style learned conf. [7]	.856	.934	.912	.859	.957	.904
Two-engine voting	.609	.772	.976	.759	.862	.796
Re-OCR: TrOCR-base [4]	.605	.811	**.979**	.758	.952	.821
Re-OCR: Florence-2 [5]	.647	.615	.478	.369	.823	.586
Re-OCR: PARSeq [3]	.946	.973	.742	.879	**.993**	.907
Re-OCR: PARSeq, graded	.946	.973	.742	.879	**.993**	.907
VLM judge: Qwen2.5-VL-7B [6]	.760	.891	.866	.700	.888	.821
VLM judge, oracle threshold†	.762	.902	.883	.719	.928	.839
Best baseline per dataset, chosen in hindsight‡	.946	.973	.979	.879	.993	.954
Strong learned baselines on our evidence						
LR on flattened graph evidence	.949	.977	.978	.934	.976	.963
LR on character evidence	.948	.980	**.979**	.928	.976	.962
GBDT on all evidence	**.951**	.980	.978	.944	.989	.968
DeepSets over rivals (no edges)	.949	**.983**	.978	.947	.982	.968
LOCUS-GNN from scratch	**.951**	**.983**	**.979**	**.950**	.987	**.970**
Ours						
LOCUS zero-shot (no labels)	.918	.904	.798	.825	.968	.883
LOCUS (k-shot)	.950	.981	**.979**	.945	.985	.968

Table S13. AUROC at $k = 10$.

Method	FUNSD	CORD	SROIE	IAM	IIIT5K	Avg.
Baselines						
Engine MSP confidence	.861	.892	.859	.878	.938	.886
ConfidNet-style learned conf. [7]	.763	.907	.867	.849	.981	.873
Two-engine voting	.680	.807	.976	.767	.905	.827
Re-OCR: TrOCR-base [4]	.675	.841	.978	.766	.966	.845
Re-OCR: Florence-2 [5]	.715	.685	.578	.522	.879	.676
Re-OCR: PARSeq [3]	.939	.975	.763	.885	.991	.911
Re-OCR: PARSeq, graded	**.976**	.986	.760	.914	**.995**	.926
VLM judge: Qwen2.5-VL-7B [6]	.846	.945	.941	.788	.974	.899
VLM judge, oracle threshold†	.846	.945	.941	.788	.974	.899
Best baseline per dataset, chosen in hindsight‡	.976	.986	.978	.914	.995	.970
Strong learned baselines on our evidence						
LR on flattened graph evidence	.972	**.995**	**.991**	.961	.991	**.982**
LR on character evidence	.961	.992	.990	.925	.991	.972
GBDT on all evidence	.952	.988	.986	.948	.982	.971
DeepSets over rivals (no edges)	.960	**.995**	.985	.960	.988	.978
LOCUS-GNN from scratch	.955	.992	.977	.957	.987	.974
Ours						
LOCUS zero-shot (no labels)	.949	.977	.966	.909	.983	.957
LOCUS (k-shot)	.967	.993	.989	**.963**	.989	.980

Table S14. AUROC at $k = 100$.

Method	FUNSD	CORD	SROIE	IAM	IIIT5K	Avg.
Baselines						
Engine MSP confidence	.861	.892	.859	.878	.938	.886
ConfidNet-style learned conf. [7]	.907	.964	.948	.894	.985	.940
Two-engine voting	.680	.807	.976	.767	.905	.827
Re-OCR: TrOCR-base [4]	.675	.841	.978	.766	.966	.845
Re-OCR: Florence-2 [5]	.715	.685	.578	.522	.879	.676
Re-OCR: PARSeq [3]	.939	.975	.763	.885	.991	.911
Re-OCR: PARSeq, graded	**.976**	.986	.760	.914	.995	.926
VLM judge: Qwen2.5-VL-7B [6]	.846	.945	.941	.788	.974	.899
VLM judge, oracle threshold†	.846	.945	.941	.788	.974	.899
Best baseline per dataset, chosen in hindsight‡	.976	.986	.978	.914	.995	.970
Strong learned baselines on our evidence						
LR on flattened graph evidence	.975	**.996**	.994	.976	.993	**.987**
LR on character evidence	.973	.994	**.995**	.974	.994	.986
GBDT on all evidence	.975	.995	**.995**	.976	**.996**	**.987**
DeepSets over rivals (no edges)	.972	.995	.993	**.979**	.995	**.987**
LOCUS-GNN from scratch	.970	.994	.993	.977	.995	.986
Ours						
LOCUS zero-shot (no labels)	.949	.977	.966	.909	.983	.957
LOCUS (k-shot)	.975	.993	.993	.978	.994	**.987**

Table S15. AURC at $k = 25$ (lower is better; best = lowest).

Method	FUNSD	CORD	SROIE	IAM	IIIT5K	Avg.
Baselines						
Engine MSP confidence	.144	.140	.211	.203	.069	.153
ConfidNet-style learned conf. [7]	.164	.124	.184	.226	.054	.150
Two-engine voting	.220	.170	.152	.272	.076	.178
Re-OCR: TrOCR-base [4]	.229	.154	.151	.273	.058	.173
Re-OCR: Florence-2 [5]	.198	.233	.377	.424	.083	.263
Re-OCR: PARSeq [3]	.125	.109	.246	.186	.050	.143
Re-OCR: PARSeq, graded	**.090**	.099	.242	.168	**.049**	.130
VLM judge: Qwen2.5-VL-7B [6]	.153	.127	.172	.250	.056	.151
VLM judge, oracle threshold†	.153	.127	.172	.250	.056	.151
Best baseline per dataset, chosen in hindsight‡	.090	.099	.151	.168	.049	.111
Strong learned baselines on our evidence						
LR on flattened graph evidence	.092	**.096**	.139	**.146**	.052	**.105**
LR on character evidence	.094	.098	.139	.152	.050	.107
GBDT on all evidence	.098	.100	**.138**	.150	.052	.108
DeepSets over rivals (no edges)	.094	.097	**.138**	.147	.052	.106
LOCUS-GNN from scratch	.095	.098	.139	.149	.052	.107
Ours						
LOCUS zero-shot (no labels)	.110	.104	.152	.180	.054	.120
LOCUS (k-shot)	.092	.097	.139	.149	.052	.106

Table S16. Coverage at 95% accepted precision at $k = 25$ (higher is better).

Method	FUNSD	CORD	SROIE	IAM	IIIT5K	Avg.
Baselines						
Engine MSP confidence	.346	.408	.091	.091	.631	.313
ConfidNet-style learned conf. [7]	.053	.465	.233	.007	.726	.297
Two-engine voting	.021	.409	.547	.001	.635	.322
Re-OCR: TrOCR-base [4]	.000	.447	.550	.001	.715	.343
Re-OCR: Florence-2 [5]	.311	.249	.092	.001	.595	.249
Re-OCR: PARSeq [3]	.006	.615	.305	.455	.746	.426
Re-OCR: PARSeq, graded	**.646**	.614	.304	.454	**.747**	.553
VLM judge: Qwen2.5-VL-7B [6]	.147	.351	.238	.052	.716	.301
VLM judge, oracle threshold†	.147	.351	.238	.052	.716	.301
Best baseline per dataset, chosen in hindsight‡	.646	.615	.550	.455	.747	.603
Strong learned baselines on our evidence						
LR on flattened graph evidence	.645	.625	.546	.483	.745	.609
LR on character evidence	.631	.624	.545	.446	.745	.598
GBDT on all evidence	.625	.621	.548	.467	.746	.601
DeepSets over rivals (no edges)	.639	**.627**	.552	**.497**	**.747**	**.612**
LOCUS-GNN from scratch	.630	.624	.550	.482	**.747**	.607
Ours						
LOCUS zero-shot (no labels)	.270	.551	.401	.346	.739	.461
LOCUS (k-shot)	.628	.626	**.553**	.486	.744	.607

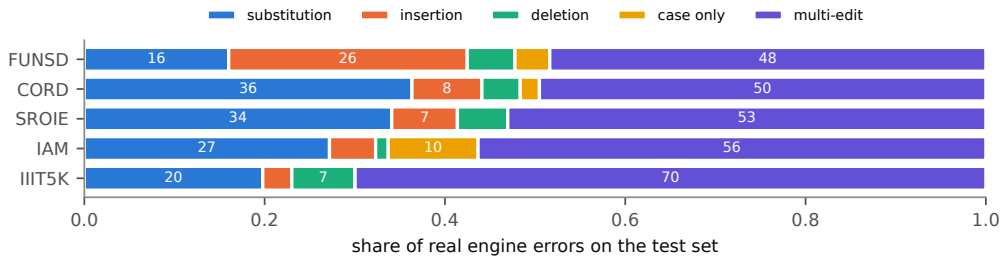


Figure S3. Types of real engine errors on each test set (share of errors; labels for shares above 7%). Handwriting and receipt lines are dominated by multi-edit errors, which bounds what a single-edit neighbourhood can repair (Table S33).

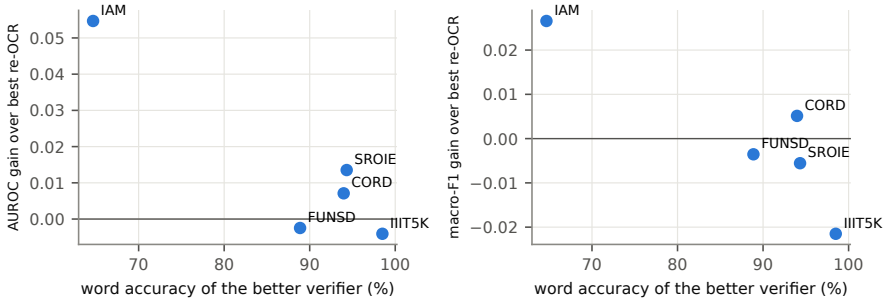


Figure S4. Gain of LOCUS over the best re-recognition baseline against the word accuracy of the better verifier (one point per dataset).

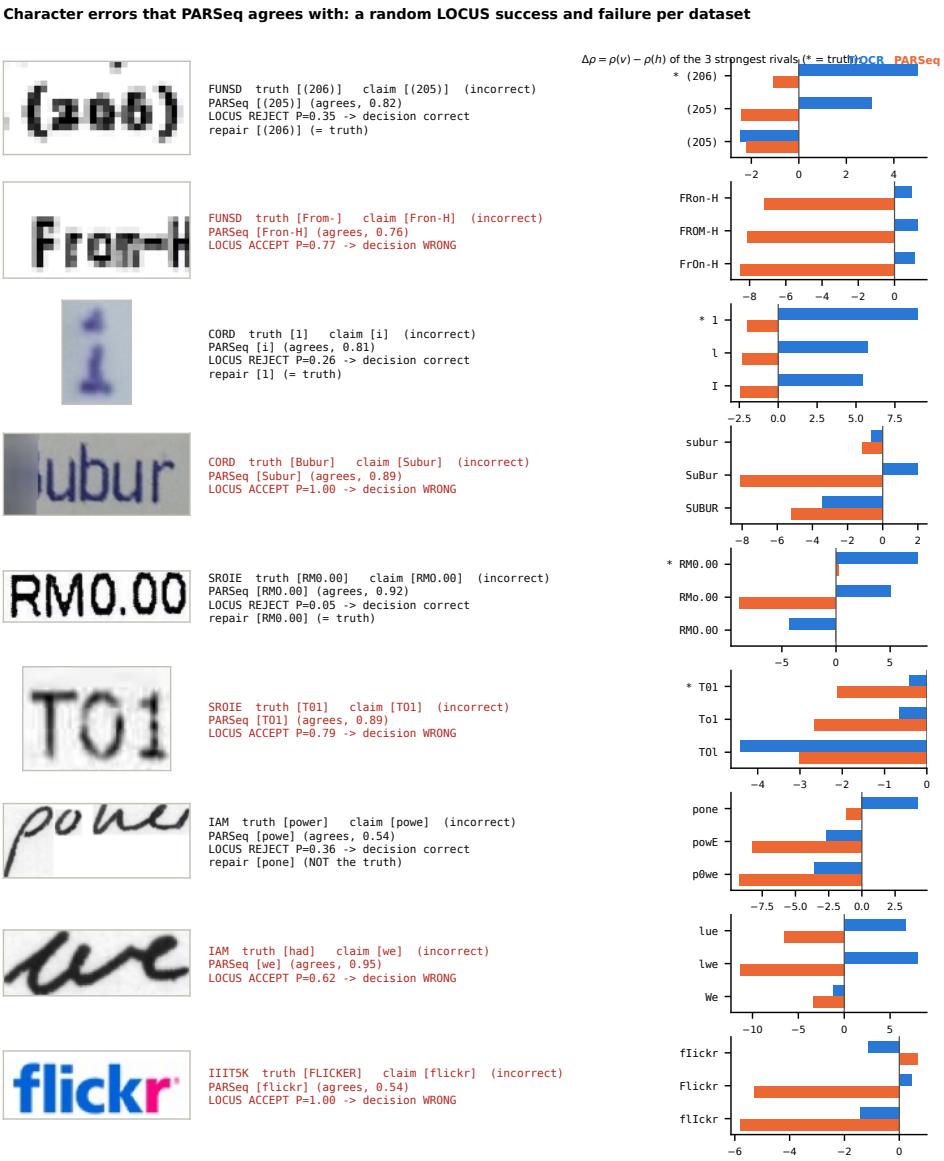


Figure S5. Errors that PARSeq re-recognition accepts (it reads the same string as the engine), restricted to genuine character errors (case-only and punctuation-only differences excluded): for each dataset one random LOCUS success and one random failure. LOCUS rejects only a small fraction of this stratum (Table S40); the manual audit (Table S41) shows why: most of these "errors" are annotation or cropping problems, i.e. the claim matches what is visible.

Table S17. Single-verifier setting (no agreement features), $k = 25$, macro-F1: effect of render normalisation when only one recogniser is available.

Variant	FUNSD	CORD	SROIE	IAM	IIIT5K
TrOCR only	.930	**.973**	**.956**	.839	**.957**
TrOCR only, w/o render-norm.	.929	.967	.902	.849	.951
PARSeq only	.932	.968	.918	**.885**	.943
PARSeq only, w/o render-norm.	**.933**	.969	.916	.884	.947

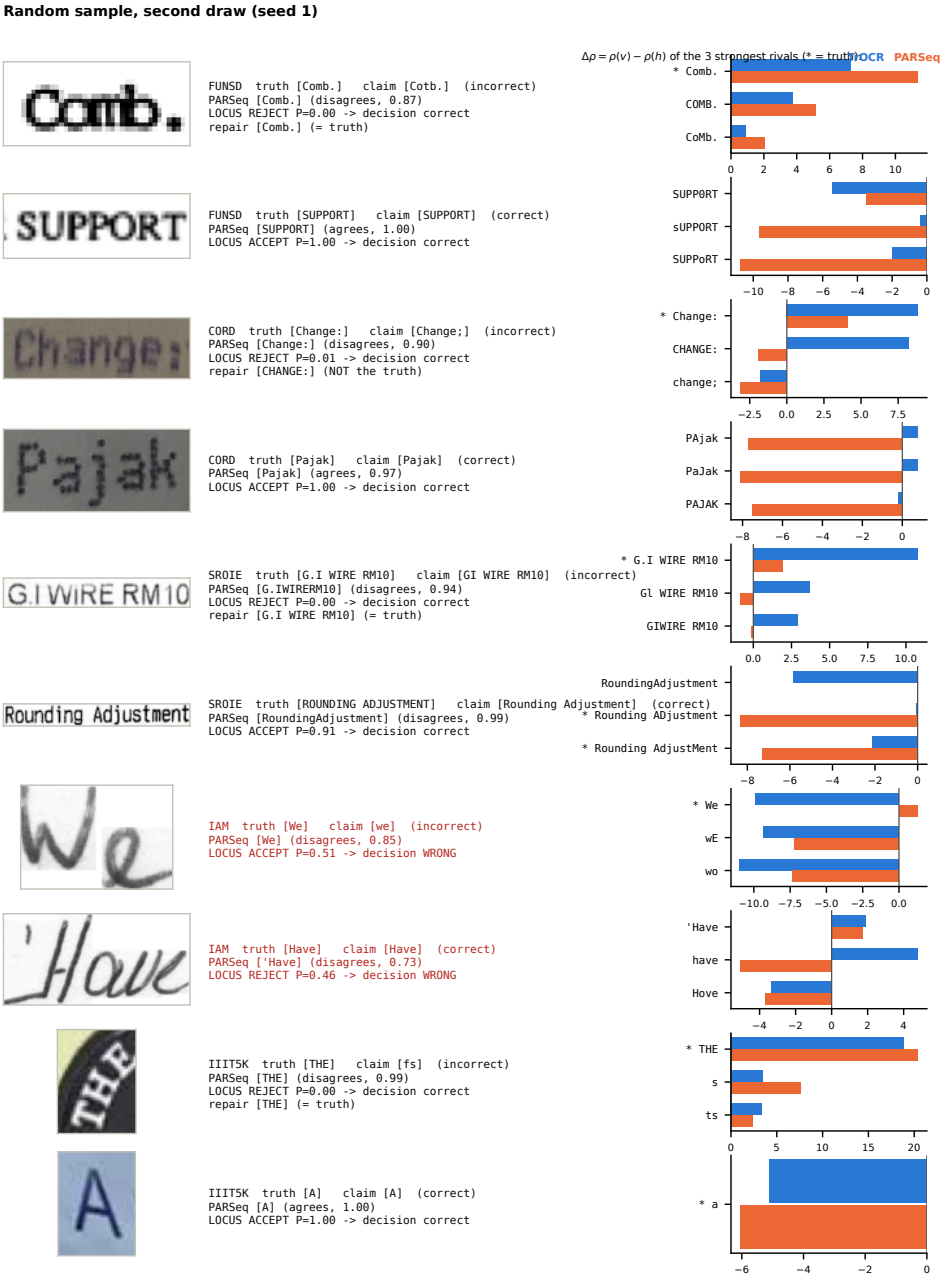


Figure S6. Second random sample (seed 1), same protocol as Fig. S13.

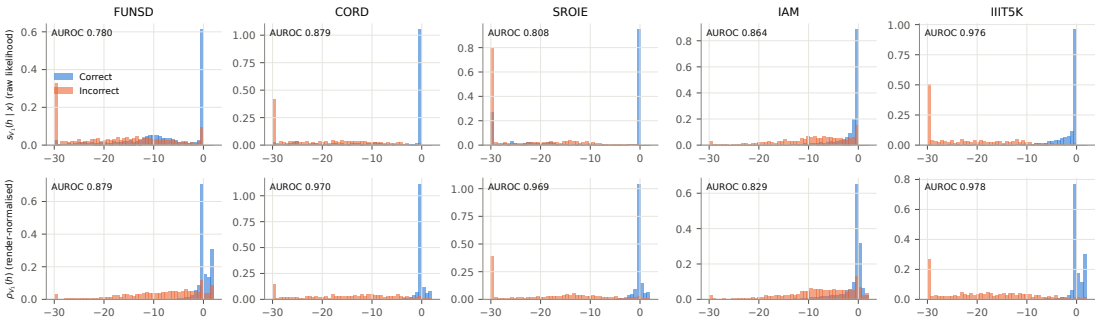


Figure S7. Render normalisation on real data. Distribution of the claim's raw TrOCR likelihood (top) and of its render-normalised ratio ρ (bottom) for correct and incorrect claims, with the AUROC of each single score.

Table S18. FUNSD: macro-f1 as a function of k (labelled words per class). Ensemble of five seeds; subscript: std. of single-seed runs.

Method	k=5	k=10	k=25	k=50	k=100	full
Engine MSP conf.	.726	.722	.756	.764	.771	.787
ConfidNet-style	.683	.702	.771	.812	.830	.856
Flat-LR	$.897_{.041}$	$.931_{.032}$	$\mathbf{.948}_{.009}$	$.946_{.007}$	$.945_{.011}$	$.949_{.000}$
CharEv-LR	$.870_{.033}$	$.915_{.039}$	$.935_{.005}$	$.937_{.001}$	$.940_{.002}$	$.948_{.001}$
GBDT	$.874_{.061}$	$.885_{.070}$	$.922_{.034}$	$.943_{.006}$	$.944_{.008}$	$\mathbf{.951}_{.001}$
DeepSets	$.898_{.021}$	$.931_{.063}$	$.942_{.042}$	$\mathbf{.947}_{.009}$	$.942_{.016}$	$.949_{.001}$
LOCUS-GNN scratch	$.868_{.033}$	$.920_{.060}$	$.933_{.031}$	$.944_{.014}$	$.938_{.022}$	$\mathbf{.951}_{.001}$
LOCUS probe	$.921_{.013}$	$.938_{.009}$	$.946_{.014}$	$\mathbf{.947}_{.003}$	$\mathbf{.947}_{.007}$	$.950_{.001}$
LOCUS + CV head	$.929_{.011}$	$.934_{.008}$	$.942_{.023}$	$.946_{.004}$	$.945_{.010}$	$.949_{.001}$
Priors-only LR	$\mathbf{.940}_{.005}$	$\mathbf{.946}_{.002}$	$.947_{.001}$	$\mathbf{.947}_{.001}$	$\mathbf{.947}_{.000}$	$.946_{.000}$
LOCUS	$.923_{.020}$	$.936_{.030}$	$.943_{.007}$	$.943_{.009}$	$.946_{.004}$	$.950_{.002}$

Table S19. CORD: macro-f1 as a function of k (labelled words per class). Ensemble of five seeds; subscript: std. of single-seed runs.

Method	k=5	k=10	k=25	k=50	k=100	full
Engine MSP conf.	.742	.775	.777	.793	.799	.805
ConfidNet-style	.751	.813	.842	.882	.905	.934
Flat-LR	$.933_{.019}$	$.957_{.018}$	$.955_{.018}$	$.963_{.012}$	$\mathbf{.980}_{.004}$	$.977_{.001}$
CharEv-LR	$.940_{.044}$	$.970_{.016}$	$.971_{.016}$	$.975_{.004}$	$.979_{.003}$	$.980_{.000}$
GBDT	$.917_{.064}$	$.960_{.016}$	$.961_{.015}$	$.970_{.008}$	$\mathbf{.980}_{.004}$	$.980_{.001}$
DeepSets	$.949_{.033}$	$.978_{.019}$	$.977_{.015}$	$.980_{.002}$	$\mathbf{.980}_{.003}$	$\mathbf{.983}_{.001}$
LOCUS-GNN scratch	$.916_{.013}$	$.969_{.025}$	$.969_{.020}$	$.977_{.002}$	$.978_{.003}$	$\mathbf{.983}_{.001}$
LOCUS probe	$.948_{.024}$	$.953_{.019}$	$.961_{.020}$	$\mathbf{.981}_{.006}$	$\mathbf{.980}_{.002}$	$.981_{.001}$
LOCUS + CV head	$.926_{.018}$	$.942_{.025}$	$.962_{.023}$	$.978_{.008}$	$.979_{.002}$	$.981_{.000}$
Priors-only LR	$\mathbf{.978}_{.002}$	$.978_{.000}$	$\mathbf{.978}_{.000}$	$.978_{.000}$	$.978_{.000}$	$.978_{.000}$
LOCUS	$.943_{.019}$	$\mathbf{.980}_{.024}$	$\mathbf{.978}_{.023}$	$\mathbf{.981}_{.002}$	$\mathbf{.980}_{.002}$	$.981_{.003}$

Table S20. SROIE: macro-f1 as a function of k (labelled words per class). Ensemble of five seeds; subscript: std. of single-seed runs.

Method	k=5	k=10	k=25	k=50	k=100	full
Engine MSP conf.	.758	.749	.780	.777	.777	.790
ConfidNet-style	.746	.785	.826	.863	.873	.912
Flat-LR	$.932_{.026}$	$.939_{.011}$	$.955_{.018}$	$.978_{.020}$	$.978_{.007}$	$.978_{.000}$
CharEv-LR	$.941_{.084}$	$.949_{.022}$	$.959_{.013}$	$.973_{.006}$	$.977_{.004}$	$\mathbf{.979}_{.000}$
GBDT	$.956_{.019}$	$.967_{.029}$	$.962_{.035}$	$.978_{.015}$	$\mathbf{.979}_{.003}$	$.978_{.001}$
DeepSets	$.924_{.029}$	$.930_{.025}$	$.968_{.013}$	$.969_{.007}$	$.976_{.005}$	$.978_{.000}$
LOCUS-GNN scratch	$.903_{.042}$	$.913_{.018}$	$.961_{.018}$	$.970_{.008}$	$.977_{.009}$	$\mathbf{.979}_{.001}$
LOCUS probe	$.961_{.028}$	$.957_{.004}$	$.973_{.007}$	$.973_{.006}$	$.975_{.004}$	$\mathbf{.979}_{.000}$
LOCUS + CV head	$.884_{.051}$	$.956_{.020}$	$.964_{.015}$	$.975_{.004}$	$.977_{.004}$	$.978_{.000}$
Priors-only LR	$\mathbf{.978}_{.034}$	$\mathbf{.978}_{.000}$	$\mathbf{.978}_{.004}$	$\mathbf{.979}_{.003}$	$\mathbf{.979}_{.000}$	$\mathbf{.979}_{.000}$
LOCUS	$.941_{.046}$	$.954_{.017}$	$.973_{.007}$	$.973_{.003}$	$.974_{.003}$	$\mathbf{.979}_{.001}$

Table S21. IAM: macro-f1 as a function of k (labelled words per class). Ensemble of five seeds; subscript: std. of single-seed runs.

Method	k=5	k=10	k=25	k=50	k=100	full
Engine MSP conf.	.700	.782	.802	.801	.801	.807
ConfidNet-style	.762	.783	.801	.812	.832	.859
Flat-LR	$.863_{.038}$	$.881_{.012}$	$.904_{.017}$	$.915_{.009}$	$.918_{.003}$	$.934_{.001}$
CharEv-LR	$.838_{.020}$	$.839_{.015}$	$.887_{.016}$	$.902_{.007}$	$.920_{.008}$	$.928_{.001}$
GBDT	$.840_{.025}$	$.882_{.053}$	$.893_{.030}$	$.920_{.013}$	$.923_{.019}$	$.944_{.002}$
DeepSets	$.873_{.054}$	$.896_{.031}$	$\mathbf{.909}_{.019}$	$.921_{.009}$	$.926_{.009}$	$.947_{.004}$
LOCUS-GNN scratch	$.865_{.039}$	$.877_{.019}$	$.908_{.023}$	$.917_{.012}$	$.927_{.008}$	$\mathbf{.950}_{.004}$
LOCUS probe	$.870_{.037}$	$.875_{.025}$	$.904_{.016}$	$.922_{.008}$	$\mathbf{.931}_{.007}$	$.944_{.003}$
LOCUS + CV head	$.851_{.026}$	$.888_{.030}$	$.905_{.009}$	$.916_{.009}$	$.925_{.009}$	$.944_{.003}$
Priors-only LR	$\mathbf{.874}_{.017}$	$\mathbf{.897}_{.013}$	$.898_{.013}$	$.908_{.003}$	$.907_{.002}$	$.907_{.001}$
LOCUS	$.869_{.020}$	$.888_{.023}$	$.906_{.011}$	$\mathbf{.923}_{.010}$	$.927_{.006}$	$.945_{.003}$

Table S22. IIIT5K: macro-f1 as a function of k (labelled words per class). Ensemble of five seeds; subscript: std. of single-seed runs.

Method	k=5	k=10	k=25	k=50	k=100	full
Engine MSP conf.	.806	.817	.788	.817	.790	.854
ConfidNet-style	.925	.915	.922	.929	.924	.957
Flat-LR	.966$_{.021}$	.974$_{.008}$	.969$_{.005}$	.967$_{.007}$	.970$_{.009}$	.976$_{.003}$
CharEv-LR	.958$_{.022}$	.965$_{.006}$	.965$_{.008}$	.970$_{.004}$	.967$_{.008}$	.976$_{.003}$
GBDT	.941$_{.007}$	.947$_{.032}$	.955$_{.019}$	.968$_{.039}$	**.983**$_{.014}$	**.989**$_{.002}$
DeepSets	.964$_{.025}$	.964$_{.030}$	.965$_{.015}$	.968$_{.009}$	.968$_{.013}$	.982$_{.008}$
LOCUS-GNN scratch	.961$_{.020}$	.961$_{.046}$	.962$_{.022}$	.969$_{.009}$	.973$_{.009}$	.987$_{.002}$
LOCUS probe	.966$_{.010}$	.967$_{.005}$	.968$_{.008}$	.970$_{.008}$	.966$_{.011}$	.978$_{.004}$
LOCUS + CV head	.963$_{.010}$	.966$_{.008}$	.971$_{.005}$	.969$_{.011}$	.973$_{.012}$	.978$_{.003}$
Priors-only LR	**.983**$_{.005}$	**.982**$_{.004}$	**.981**$_{.002}$	**.980**$_{.003}$	.980$_{.003}$	.977$_{.003}$
LOCUS	.974$_{.003}$	.975$_{.030}$	.971$_{.017}$	.972$_{.005}$	.973$_{.010}$	.985$_{.003}$

Table S23. FUNSD: auroc as a function of k (labelled words per class). Ensemble of five seeds; subscript: std. of single-seed runs.

Method	k=5	k=10	k=25	k=50	k=100	full
Engine MSP conf.	.861	.861	.861	.861	.861	.861
ConfidNet-style	.763	.763	.845	.883	.907	.935
Flat-LR	**.965**$_{.013}$	**.972**$_{.024}$	**.975**$_{.003}$	.974$_{.005}$	.975$_{.006}$	.977$_{.000}$
CharEv-LR	.943$_{.034}$	.961$_{.035}$	.973$_{.003}$	.972$_{.004}$	.973$_{.003}$	.975$_{.000}$
GBDT	.934$_{.060}$	.952$_{.030}$	.969$_{.014}$	.974$_{.005}$	.975$_{.006}$	.979$_{.001}$
DeepSets	.959$_{.031}$	.960$_{.056}$	.971$_{.021}$	.971$_{.004}$	.972$_{.004}$	.979$_{.002}$
LOCUS-GNN scratch	.954$_{.020}$	.955$_{.047}$	.969$_{.010}$	.968$_{.008}$	.970$_{.005}$	.979$_{.001}$
LOCUS probe	.956$_{.014}$	.969$_{.013}$	**.975**$_{.003}$	**.976**$_{.005}$	**.976**$_{.006}$	**.980**$_{.001}$
LOCUS + CV head	.948$_{.005}$	.966$_{.005}$	.972$_{.004}$	.975$_{.007}$	.975$_{.008}$	.979$_{.001}$
Priors-only LR	.958$_{.002}$	.967$_{.002}$	.966$_{.004}$	.966$_{.004}$	.967$_{.002}$	.968$_{.001}$
LOCUS	.964$_{.010}$	.967$_{.011}$	.974$_{.004}$	.972$_{.007}$	.975$_{.003}$	.979$_{.001}$

Table S24. CORD: auroc as a function of k (labelled words per class). Ensemble of five seeds; subscript: std. of single-seed runs.

Method	k=5	k=10	k=25	k=50	k=100	full
Engine MSP conf.	.892	.892	.892	.892	.892	.892
ConfidNet-style	.827	.907	.929	.953	.964	.983
Flat-LR	**.994**$_{.014}$	**.995**$_{.002}$	**.995**$_{.002}$	**.995**$_{.000}$	**.996**$_{.001}$	**.997**$_{.000}$
CharEv-LR	.985$_{.015}$	.992$_{.004}$	.992$_{.001}$	.993$_{.001}$	.994$_{.001}$	.996$_{.000}$
GBDT	.970$_{.048}$	.988$_{.022}$	.988$_{.026}$	.993$_{.014}$	.995$_{.001}$	**.997**$_{.000}$
DeepSets	.992$_{.025}$	**.995**$_{.004}$	.994$_{.004}$	**.995**$_{.000}$	.995$_{.001}$	**.997**$_{.000}$
LOCUS-GNN scratch	.987$_{.021}$	.992$_{.019}$	.992$_{.015}$	**.995**$_{.001}$	.994$_{.001}$	**.997**$_{.001}$
LOCUS probe	.983$_{.020}$	.991$_{.013}$	.993$_{.013}$	.994$_{.001}$	.994$_{.002}$	.996$_{.001}$
LOCUS + CV head	.982$_{.004}$	.990$_{.008}$	.993$_{.008}$	.994$_{.001}$	.995$_{.000}$	.996$_{.000}$
Priors-only LR	**.994**$_{.000}$	.994$_{.000}$	.994$_{.000}$	.994$_{.000}$	.994$_{.000}$	.995$_{.000}$
LOCUS	.989$_{.007}$	.993$_{.005}$	.993$_{.006}$	.994$_{.001}$	.993$_{.001}$	.996$_{.001}$

Table S25. SROIE: auroc as a function of k (labelled words per class). Ensemble of five seeds; subscript: std. of single-seed runs.

Method	k=5	k=10	k=25	k=50	k=100	full
Engine MSP conf.	.859	.859	.859	.859	.859	.859
ConfidNet-style	.815	.867	.912	.941	.948	.974
Flat-LR	.986$_{.010}$	.991$_{.003}$	.990$_{.003}$	.993$_{.003}$	.994$_{.001}$	.995$_{.000}$
CharEv-LR	.991$_{.010}$	.990$_{.011}$	.990$_{.004}$	**.994**$_{.002}$	**.995**$_{.001}$	.996$_{.000}$
GBDT	.978$_{.017}$	.986$_{.028}$	.991$_{.017}$	**.994**$_{.003}$	**.995**$_{.002}$	**.997**$_{.000}$
DeepSets	.983$_{.019}$	.985$_{.013}$	.992$_{.004}$	.993$_{.002}$	.993$_{.001}$	.996$_{.000}$
LOCUS-GNN scratch	.970$_{.030}$	.977$_{.009}$	.990$_{.005}$	.992$_{.002}$	.993$_{.003}$	.995$_{.000}$
LOCUS probe	.991$_{.004}$	**.992**$_{.001}$	**.993**$_{.002}$	.993$_{.001}$	.994$_{.001}$	.996$_{.000}$
LOCUS + CV head	.976$_{.004}$	.986$_{.003}$	.990$_{.004}$	.993$_{.001}$	.993$_{.001}$	.996$_{.000}$
Priors-only LR	**.992**$_{.001}$	**.992**$_{.000}$	**.993**$_{.000}$	.993$_{.000}$	.993$_{.000}$	.993$_{.000}$
LOCUS	.984$_{.011}$	.989$_{.003}$	.992$_{.001}$	.992$_{.001}$	.993$_{.000}$	.995$_{.001}$

S7. Extended results referenced in the main article

The following tables and figures belong to the main article and were moved here to respect its page limit; the main text discusses each of them.

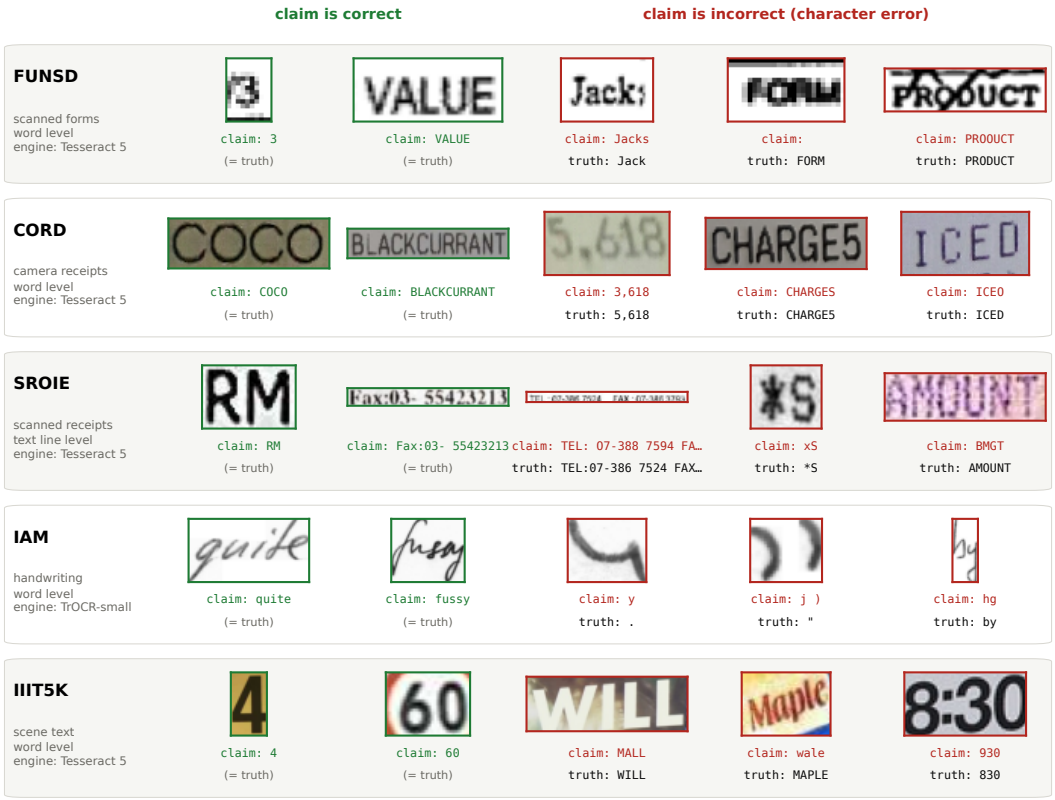


Figure S8. The five verification benchmarks. Real test crops with the claim of the main OCR engine, drawn by a seeded random choice (two correct claims and three character errors per dataset; none chosen by hand). Errors range from single confusable glyphs (`CHARGES`/`CHARGE5`, `3,618`/`5,618`) to empty or noise outputs.

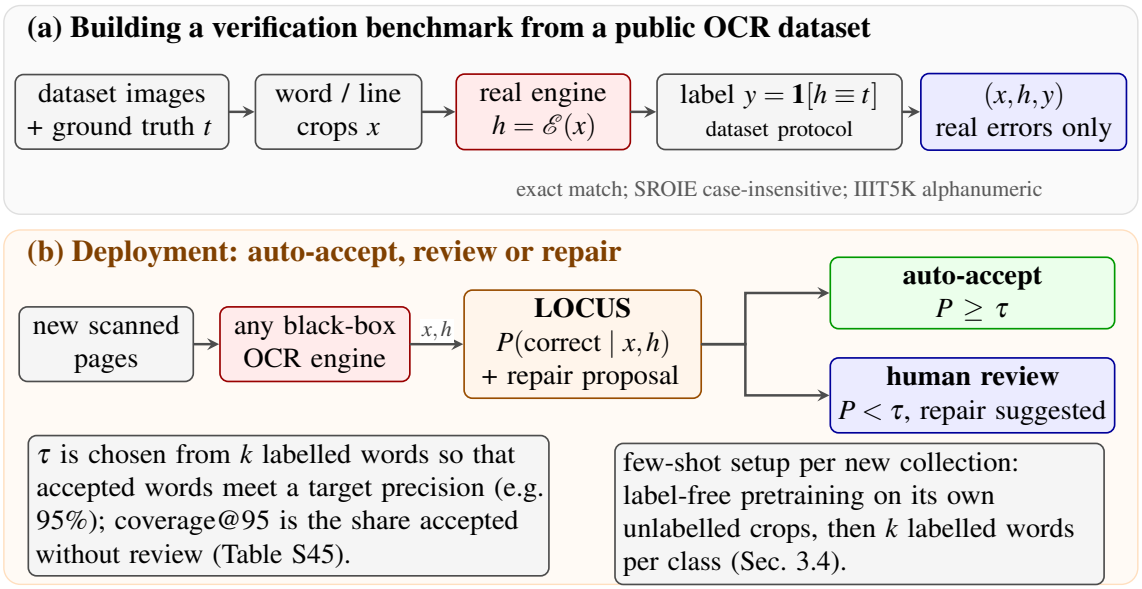


Figure S9. From benchmark to deployment. (a) Each benchmark labels real engine outputs against the dataset's own ground truth under its protocol. (b) In a digitisation pipeline, words above a precision-targeted threshold are accepted automatically and the rest go to a reviewer with a repair suggestion.

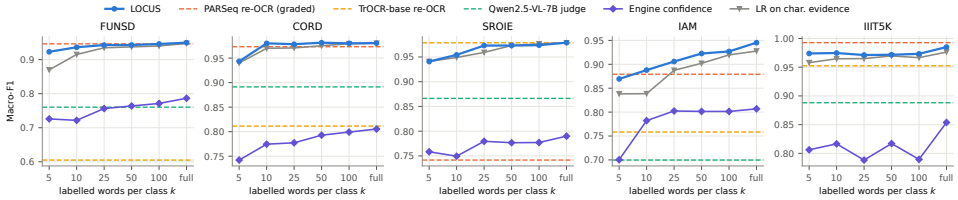


Figure S10. Label efficiency. Macro-F1 versus labelled words per class (ensembles of five seeds; dashed: label-free baselines). AUROC curves: Figure S2.

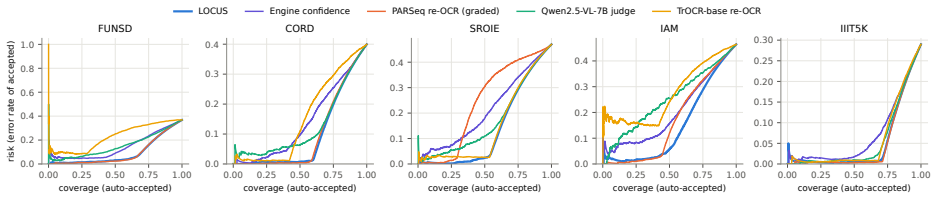


Figure S11. Risk–coverage. Error rate among auto-accepted words as a function of the fraction accepted ($k = 25$; lower is better).

Table S26. IAM: auroc as a function of k (labelled words per class). Ensemble of five seeds; subscript: std. of single-seed runs.

Method	k=5	k=10	k=25	k=50	k=100	full
Engine MSP conf.	.878	.878	.878	.878	.878	.878
ConfidNet-style	.835	.849	.861	.876	.894	.917
Flat-LR	.961$_{.010}$	.961$_{.008}$	.970$_{.008}$	.973$_{.006}$	.976$_{.002}$	.982$_{.000}$
CharEv-LR	.927$_{.011}$	.925$_{.011}$	.959$_{.011}$	.970$_{.002}$	.974$_{.003}$	.978$_{.000}$
GBDT	.929$_{.027}$	.948$_{.081}$	.964$_{.012}$	.973$_{.002}$	.976$_{.006}$	**.986**$_{.000}$
DeepSets	**.969**$_{.012}$	.960$_{.021}$	**.972**$_{.014}$	.975$_{.010}$	**.979**$_{.005}$	**.986**$_{.001}$
LOCUS-GNN scratch	.958$_{.016}$	.957$_{.014}$	.969$_{.013}$	.972$_{.009}$	.977$_{.008}$	**.986**$_{.001}$
LOCUS probe	.953$_{.014}$	.957$_{.014}$	.970$_{.007}$	**.976**$_{.001}$	**.979**$_{.001}$	.985$_{.001}$
LOCUS + CV head	.946$_{.020}$	.960$_{.022}$	.969$_{.003}$	.973$_{.004}$	.978$_{.003}$	.985$_{.001}$
Priors-only LR	.950$_{.011}$	**.964**$_{.006}$	.963$_{.005}$	.965$_{.001}$	.965$_{.000}$	.965$_{.000}$
LOCUS	.953$_{.018}$	.963$_{.016}$	.969$_{.007}$	.974$_{.005}$	.978$_{.004}$	**.986**$_{.000}$

Table S27. IIIT5K: auroc as a function of k (labelled words per class). Ensemble of five seeds; subscript: std. of single-seed runs.

Method	k=5	k=10	k=25	k=50	k=100	full
Engine MSP conf.	.938	.938	.938	.938	.938	.938
ConfidNet-style	.982	.981	.980	.984	.985	.988
Flat-LR	.990$_{.002}$	.991$_{.001}$	.991$_{.002}$	.992$_{.002}$	.993$_{.001}$	.994$_{.000}$
CharEv-LR	.990$_{.002}$	.991$_{.000}$	.993$_{.006}$	**.994**$_{.002}$	.994$_{.002}$	**.997**$_{.000}$
GBDT	.976$_{.012}$	.982$_{.012}$	.990$_{.026}$	**.994**$_{.003}$	**.996**$_{.000}$	.995$_{.000}$
DeepSets	.990$_{.003}$	.988$_{.007}$	.990$_{.005}$	**.994**$_{.005}$	.995$_{.003}$	.995$_{.001}$
LOCUS-GNN scratch	.987$_{.004}$	.987$_{.007}$	.990$_{.002}$	**.994**$_{.001}$	.995$_{.001}$	.995$_{.001}$
LOCUS probe	**.991**$_{.002}$	**.992**$_{.001}$	.993$_{.001}$	.993$_{.001}$	.995$_{.002}$	.996$_{.000}$
LOCUS + CV head	.984$_{.002}$	.985$_{.006}$	**.994**$_{.002}$	**.994**$_{.002}$	.995$_{.001}$	.995$_{.001}$
Priors-only LR	**.991**$_{.000}$	.991$_{.000}$	.991$_{.000}$	.991$_{.000}$	.991$_{.000}$	.991$_{.001}$
LOCUS	.986$_{.002}$	.989$_{.002}$	.991$_{.002}$	.992$_{.001}$	.994$_{.002}$	.995$_{.001}$

Table S28. All metrics on FUNSD at $k = 25$ (ensemble of five seeds for learned methods; mean over seeds for threshold baselines). †oracle threshold tuned on test.

Method	Macro-F1	F1$_{err}$	Acc.	AUROC	AUPRC$_{err}$	AURC↓	Cov@95
Engine MSP confidence	.756	.734	.759	.861	.750	.144	.346
ConfidNet-style learned conf. [7]	.771	.723	.781	.845	.805	.164	.053
Two-engine voting	.609	.641	.612	.680	.479	.220	.021
Re-OCR: TrOCR-base [4]	.605	.636	.608	.675	.475	.229	.000
Re-OCR: Florence-2 [5]	.647	.670	.649	.715	.508	.198	.311
Re-OCR: PARSeq [3]	.946	.931	.951	.939	.907	.125	.006
Re-OCR: PARSeq, graded	.946	.931	.951	**.976**	.970	**.090**	**.646**
VLM judge: Qwen2.5-VL-7B [6]	.760	.683	.785	.846	.764	.153	.147
VLM judge, oracle threshold†	.762	.684	.789	.846	.764	.153	.147
LR on flattened graph evidence	**.948**	**.933**	**.952**	.975	**.973**	.092	.645
LR on character evidence	.935	.916	.940	.973	.969	.094	.631
GBDT on all evidence	.922	.903	.927	.969	.964	.098	.625
DeepSets over rivals (no edges)	.942	.926	.947	.971	.968	.094	.639
LOCUS-GNN from scratch	.933	.914	.938	.969	.966	.095	.630
LOCUS zero-shot (no labels)	.918	.890	.927	.949	.953	.110	.270
LOCUS (k-shot)	.943	.926	.948	.974	.971	.092	.628
LOCUS + CV-selected head	.942	.926	.947	.972	.971	.094	.642
LOCUS linear probe	.946	.930	.950	.975	**.973**	.093	.639
Fusion of label-free priors (LR)	.947	.932	**.952**	.966	.968	.101	.643
LOCUS rule (zero-shot, ρ margin)	.866	.836	.873	.943	.939	.110	.471

Table S29. All metrics on CORD at $k = 25$ (ensemble of five seeds for learned methods; mean over seeds for threshold baselines). †oracle threshold tuned on test.

Method	Macro-F1	F1$_{err}$	Acc.	AUROC	AUPRC$_{err}$	AURC↓	Cov@95
Engine MSP confidence	.777	.759	.781	.892	.839	.140	.408
ConfidNet-style learned conf. [7]	.842	.818	.846	.929	.897	.124	.465
Two-engine voting	.772	.775	.772	.807	.635	.170	.409
Re-OCR: TrOCR-base [4]	.811	.807	.812	.841	.679	.154	.447
Re-OCR: Florence-2 [5]	.615	.677	.625	.685	.514	.233	.249
Re-OCR: PARSeq [3]	.973	.968	.974	.975	.945	.109	.615
Re-OCR: PARSeq, graded	.973	.968	.974	.986	.970	.099	.614
VLM judge: Qwen2.5-VL-7B [6]	.891	.862	.899	.945	.943	.127	.351
VLM judge, oracle threshold†	.902	.879	.908	.945	.943	.127	.351
LR on flattened graph evidence	.955	.945	.958	**.995**	**.995**	**.096**	.625
LR on character evidence	.971	.965	.972	.992	.991	.098	.624
GBDT on all evidence	.961	.953	.963	.988	.981	.100	.621
DeepSets over rivals (no edges)	.977	.972	.978	.994	.994	.097	**.627**
LOCUS-GNN from scratch	.969	.963	.970	.992	.991	.098	.624
LOCUS zero-shot (no labels)	.904	.887	.907	.977	.972	.104	.551
LOCUS (k-shot)	**.978**	**.974**	**.979**	.993	.992	.097	.626
LOCUS + CV-selected head	.962	.953	.963	.993	.993	.097	.623
LOCUS linear probe	.961	.952	.963	.993	.992	.099	.625
Fusion of label-free priors (LR)	**.978**	**.974**	**.979**	.994	.994	.097	.626
LOCUS rule (zero-shot, ρ margin)	.934	.923	.935	.991	.990	.098	.618

Table S30. All metrics on SROIE at $k = 25$ (ensemble of five seeds for learned methods; mean over seeds for threshold baselines). †oracle threshold tuned on test.

Method	Macro-F1	$F1_{err}$	Acc.	AUROC	$AUPRC_{err}$	AURC↓	Cov@95
Engine MSP confidence	.780	.766	.781	.859	.826	.211	.091
ConfidNet-style learned conf. [7]	.826	.822	.826	.912	.903	.184	.233
Two-engine voting	.976	.975	.976	.976	.963	.152	.547
Re-OCR: TrOCR-base [4]	**.979**	**.977**	**.979**	.978	.969	.151	.550
Re-OCR: Florence-2 [5]	.478	.676	.553	.578	.512	.377	.092
Re-OCR: PARSeq [3]	.742	.787	.750	.763	.653	.246	.305
Re-OCR: PARSeq, graded	.742	.787	.750	.760	.681	.242	.304
VLM judge: Qwen2.5-VL-7B [6]	.866	.845	.870	.941	.948	.172	.238
VLM judge, oracle threshold†	.883	.870	.885	.941	.948	.172	.238
LR on flattened graph evidence	.955	.951	.956	.990	.991	.139	.546
LR on character evidence	.959	.956	.960	.990	.991	.139	.545
GBDT on all evidence	.962	.959	.962	.991	.991	**.138**	.548
DeepSets over rivals (no edges)	.968	.965	.968	.992	.993	**.138**	.552
LOCUS-GNN from scratch	.961	.958	.962	.990	.992	.139	.550
LOCUS zero-shot (no labels)	.798	.823	.801	.966	.968	.152	.401
LOCUS (k-shot)	.973	.971	.973	.992	.993	.139	.553
LOCUS + CV-selected head	.964	.961	.964	.990	.991	.139	.548
LOCUS linear probe	.973	.971	.973	**.993**	**.994**	**.138**	**.555**
Fusion of label-free priors (LR)	.978	**.977**	**.979**	**.993**	**.994**	**.138**	.553
LOCUS rule (zero-shot, ρ margin)	.910	.910	.911	.981	.981	.144	.507

Table S31. All metrics on IAM at $k = 25$ (ensemble of five seeds for learned methods; mean over seeds for threshold baselines). †oracle threshold tuned on test.

Method	Macro-F1	$F1_{err}$	Acc.	AUROC	$AUPRC_{err}$	AURC↓	Cov@95
Engine MSP confidence	.802	.791	.804	.878	.859	.203	.091
ConfidNet-style learned conf. [7]	.801	.784	.803	.861	.865	.226	.007
Two-engine voting	.759	.769	.760	.767	.663	.272	.001
Re-OCR: TrOCR-base [4]	.758	.769	.759	.766	.661	.273	.001
Re-OCR: Florence-2 [5]	.369	.643	.488	.522	.475	.424	.001
Re-OCR: PARSeq [3]	.879	.881	.879	.885	.799	.186	.455
Re-OCR: PARSeq, graded	.879	.881	.879	.914	.877	.168	.454
VLM judge: Qwen2.5-VL-7B [6]	.700	.713	.700	.788	.765	.250	.052
VLM judge, oracle threshold†	.719	.694	.722	.788	.765	.250	.052
LR on flattened graph evidence	.904	.895	.905	.970	.968	**.146**	.483
LR on character evidence	.887	.878	.888	.959	.956	.152	.446
GBDT on all evidence	.893	.884	.894	.964	.961	.150	.467
DeepSets over rivals (no edges)	**.909**	**.901**	**.910**	**.972**	**.971**	.147	**.497**
LOCUS-GNN from scratch	.908	.899	.909	.969	.970	.149	.482
LOCUS zero-shot (no labels)	.825	.814	.826	.909	.895	.180	.346
LOCUS (k-shot)	.906	.896	.907	.969	.968	.149	.486
LOCUS + CV-selected head	.905	.898	.906	.969	.967	**.146**	.490
LOCUS linear probe	.904	.895	.905	.970	.970	.148	.473
Fusion of label-free priors (LR)	.898	.891	.899	.963	.959	.149	.481
LOCUS rule (zero-shot, ρ margin)	.844	.838	.844	.933	.934	.167	.330

Table S32. All metrics on IIIT5K at $k = 25$ (ensemble of five seeds for learned methods; mean over seeds for threshold baselines). †oracle threshold tuned on test.

Method	Macro-F1	$F1_{err}$	Acc.	AUROC	$AUPRC_{err}$	AURC↓	Cov@95
Engine MSP confidence	.788	.738	.801	.938	.857	.069	.631
ConfidNet-style learned conf. [7]	.922	.892	.933	.980	.958	.054	.726
Two-engine voting	.862	.819	.876	.905	.696	.076	.635
Re-OCR: TrOCR-base [4]	.952	.934	.960	.966	.879	.058	.715
Re-OCR: Florence-2 [5]	.823	.776	.836	.879	.635	.083	.595
Re-OCR: PARSeq [3]	**.993**	**.990**	**.994**	.991	.984	.050	.746
Re-OCR: PARSeq, graded	**.993**	**.990**	**.994**	**.995**	**.994**	**.049**	.747
VLM judge: Qwen2.5-VL-7B [6]	.888	.850	.901	.974	.943	.056	.716
VLM judge, oracle threshold†	.928	.899	.940	.974	.943	.056	.716
LR on flattened graph evidence	.969	.956	.974	.991	.988	.052	.745
LR on character evidence	.965	.951	.971	.993	.990	.050	.745
GBDT on all evidence	.955	.937	.962	.990	.987	.052	.746
DeepSets over rivals (no edges)	.965	.951	.971	.990	.987	.052	.747
LOCUS-GNN from scratch	.962	.946	.968	.990	.987	.052	.747
LOCUS zero-shot (no labels)	.968	.955	.973	.983	.954	.054	.739
LOCUS (k-shot)	.971	.959	.976	.991	.985	.052	.744
LOCUS + CV-selected head	.971	.960	.976	.994	.992	**.049**	.747
LOCUS linear probe	.968	.956	.974	.993	.991	.051	**.748**
Fusion of label-free priors (LR)	.981	.974	.985	.991	.985	.052	.747
LOCUS rule (zero-shot, ρ margin)	.790	.742	.801	.981	.963	.055	.735

Table S33. Counterfactual graphs on the test sets: average hypotheses per claim and share of real engine errors whose ground truth equals TrOCR's reading, PARSeq's reading, or *any* node of $G(h)$.

Dataset	nodes	truth $= g_1$ (%)	truth $= g_2$ (%)	truth $\in G(h)$ (%)
FUNSD	17.2	31.3	72.4	82.8
CORD	15.3	70.4	89.5	94.9
SROIE	21.2	89.5	44.1	92.6
IAM	17.2	14.4	46.5	55.4
IIIT5K	15.9	86.8	95.3	97.0

Table S34. Render normalisation by claim type: AUROC of TrOCR's raw claim likelihood $s(h|x)$ (raw) and of the render-normalised ratio $\rho(h)$ (ρ), on test claims with digits, upper-case words and lower-/mixed-case words (subsets with fewer than 50 claims omitted).

	digits		upper-case		lower/mixed	
Dataset	raw	ρ	raw	ρ	raw	ρ
FUNSD	0.932	0.911	0.950	0.951	0.714	0.807
CORD	0.981	0.981	0.984	0.988	0.752	0.948
SROIE	0.901	0.976	1.000	1.000	0.722	0.972
IAM	0.740	0.638	–	–	0.849	0.814
IIIT5K	0.940	0.903	0.986	0.985	0.964	0.977

Table S35. Transfer. Left: engine transfer (train on one engine's claims, test on the other's; in-engine: trained and tested on the target engine). Right: dataset transfer, macro-F1 (rows: training dataset, fully supervised; shaded: in-domain).

		$k = 25$			full		
Dataset	Train→Test	F1 in	F1 tr.	AUROC tr.	F1 in	F1 tr.	AUROC tr.
FUNSD	Tesseract→TrOCR-s	.955	.838	.963	.959	.958	.988
FUNSD	TrOCR-s→Tesseract	.941	.927	.964	.951	.947	.976
CORD	Tesseract→TrOCR-s	.956	.689	.974	.971	.891	.989
CORD	TrOCR-s→Tesseract	.980	.817	.982	.979	.879	.993
SROIE	Tesseract→TrOCR-s	.629	.745	.862	.716	.773	.884
SROIE	TrOCR-s→Tesseract	.973	.776	.922	.979	.787	.941
IIIT5K	Tesseract→TrOCR-s	.990	.980	.995	.984	.993	.996
IIIT5K	TrOCR-s→Tesseract	.971	.972	.988	.985	.966	.991

Train\Test	FUNSD	CORD	SROIE	IAM	IIIT5K
FUNSD	.949	.977	.861	.905	.938
CORD	.942	.981	.959	.882	.914
SROIE	.620	.823	.978	.779	.946
IAM	.880	.878	.734	.944	.925
IIIT5K	.946	.973	.767	.899	.978

Table S36. Recogniser dependence. Word accuracy (%) of each verifier's reading, the better re-recognition baseline, and LOCUS ($k = 25$) against it.

	verifier accuracy		best	macro-F1			AUROC		
Dataset	TrOCR	PARSeq	baseline	base	LOCUS	Δ	base	LOCUS	Δ
FUNSD	38.0	88.9	PARSeq	0.946	0.943	-0.004	0.976	0.974	-0.002
CORD	69.8	94.0	PARSeq	0.973	0.978	+0.005	0.986	0.993	+0.007
SROIE	94.3	49.5	TrOCR-base	0.979	0.973	-0.006	0.978	0.992	+0.014
IAM	42.4	64.7	PARSeq	0.879	0.906	+0.027	0.914	0.969	+0.055
IIIT5K	92.7	98.5	PARSeq	0.993	0.971	-0.021	0.995	0.991	-0.004

Table S37. Word-level intervals inflated by the worst-case design effect (Deff) estimated on FUNSD and CORD. sig.: 95% interval excludes zero.

Dataset	Deff	Baseline	ΔAUROC	95% CI	sig.	ΔF1	95% CI	sig.
SROIE	3.67	PARSeq re-OCR (graded)	+0.232	[+0.204, +0.260]	yes	+0.232	[+0.204, +0.259]	yes
SROIE	3.67	TrOCR-base re-OCR	+0.014	[+0.007, +0.020]	yes	-0.005	[-0.013, +0.002]	no
IAM	2.29	PARSeq re-OCR (graded)	+0.055	[+0.042, +0.067]	yes	+0.026	[+0.009, +0.044]	yes
IAM	2.29	TrOCR-base re-OCR	+0.203	[+0.185, +0.221]	yes	+0.148	[+0.127, +0.168]	yes

Table S38. Rendering sensitivity: AUROC of the claim's ratio $\rho_{V_1} + \rho_{V_2}$ alone, AUROC of the training-free ρ-margin rule, and macro-F1 of LOCUS trained on 1,000 labelled words (3 seeds), for each rendering.

	FUNSD			CORD			IAM		
Rendering	$\rho(h)$	rule	LOCUS	$\rho(h)$	rule	LOCUS	$\rho(h)$	rule	LOCUS
Liberation Sans (default)	.903	.923	.937	.981	**.996**	.972	.921	.921	.921
Liberation Serif	.906	.928	.938	.984	.995	.972	.918	**.944**	.920
Liberation Mono	.907	**.940**	.938	**.990**	.994	**.977**	.915	.931	**.923**
DejaVu Sans	**.912**	.920	.937	.984	.995	.976	.922	.923	.919
Comic Neue (handwriting-like)	.907	.928	**.939**	.989	.993	.976	.917	.937	.921
Liberation Sans, glyphs 0.6x	.903	.924	**.939**	.983	**.996**	.976	.915	.919	**.923**
Liberation Sans, glyphs 0.9x	.901	.924	.937	.980	.995	.973	**.923**	.923	.921

Table S39. AUROC on all errors / on genuine character errors only ($k = 25$). Removed: share of incorrect claims that differ from the ground truth only in case or punctuation.

Dataset	removed	LOCUS	PARSeq graded	TrOCR re-OCR	Qwen2.5-VL-7B	Engine conf.
FUNSD	51%	0.974 / 0.976	**0.976 / 0.979**	0.675 / 0.691	0.846 / 0.928	0.861 / 0.886
CORD	19%	**0.993 / 0.996**	0.986 / 0.987	0.841 / 0.845	0.945 / 0.989	0.892 / 0.907
SROIE	23%	**0.992 / 0.999**	0.760 / 0.756	0.978 / 0.991	0.941 / 0.982	0.859 / 0.886
IAM	18%	**0.969 / 0.980**	0.914 / 0.908	0.766 / 0.802	0.788 / 0.818	0.878 / 0.916
IIIT5K	0%	0.991 / 0.991	**0.995 / 0.995**	0.966 / 0.966	0.974 / 0.974	0.938 / 0.938

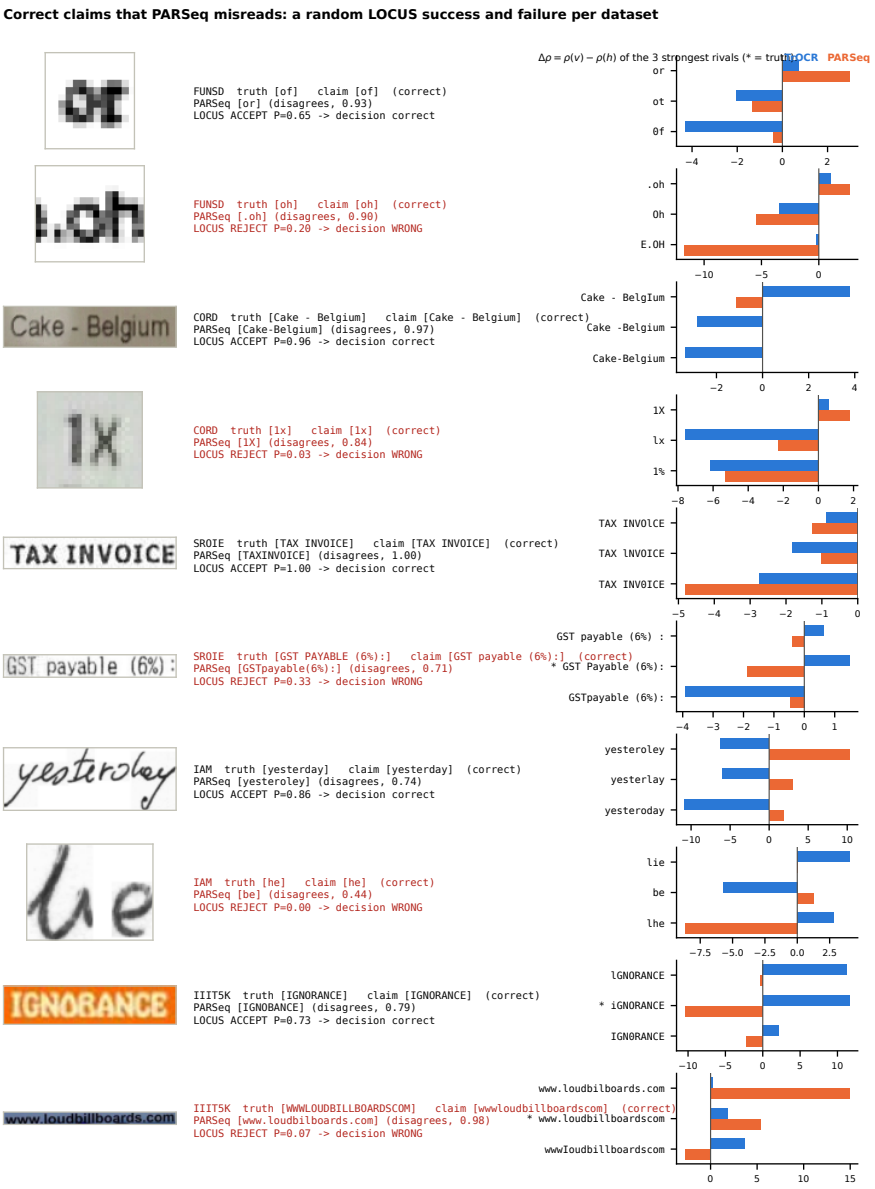


Figure S12. Correct claims that PARSeq misreads (re-recognition would reject them): one random LOCUS success and one random failure per dataset. Comparing the claim with its rivals under two verifiers helps most in this regime: a rival that one verifier prefers is typically rejected by the other.

Table S40. Regimes on the full test sets ($k = 25$). Artefacts: incorrect claims that differ from the ground truth only in case or punctuation. Right: errors PARSeq agrees with, restricted to genuine character errors.

	all errors		correct claims PARSeq misreads		character errors PARSeq agrees with	
Dataset	count	artefacts	count	LOCUS accepts	count	LOCUS rejects
FUNSD	1474	751 (51%)	38	14 (37%)	47	3 (6%)
CORD	941	181 (19%)	43	28 (65%)	9	1 (11%)
SROIE	1880	438 (23%)	971	953 (98%)	7	4 (57%)
IAM	1857	326 (18%)	419	312 (74%)	53	14 (26%)
IIIT5K	869	0 (0%)	4	3 (75%)	14	0 (0%)

Table S41. Manual audit of random test words (20 per cell, fewer where the stratum is smaller). G: genuine, N: label noise, C: crop problem, A: ambiguous, P: claim correct only under the dataset's normalisation protocol although its exact appearance differs from the image (IIIT5K only).

	LOCUS decision wrong					errors PARSeq agrees with			
Dataset	G	N	C	A	P	G	N	C	A
FUNSD	8	6	4	2	0	0	16	3	1
CORD	10	3	4	3	0	1	8	4	5
SROIE	7	12	1	0	0	5	10	1	4
IAM	16	1	0	3	0	4	1	11	4
IIIT5K	1	3	0	0	16	0	11	2	1

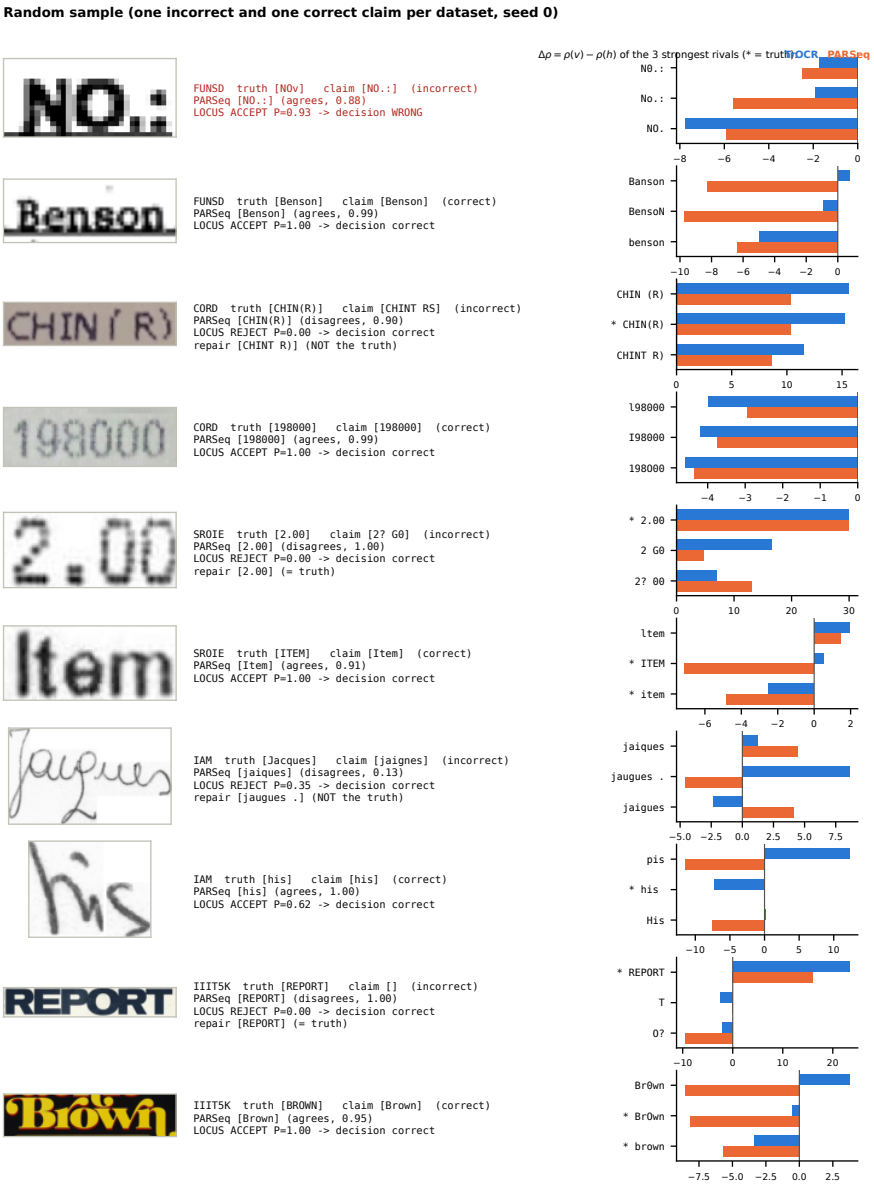


Figure S13. Random sample (seed 0): one incorrect and one correct claim per dataset, drawn without regard to LOCUS' output; wrong decisions in red. Annotation errors also appear (e.g. FUNSD ground truth "NOv" for a crop reading "NO.:"; Table S41).

Table S42. Decision accuracy (%) stratified by which verifier reads the word correctly ($k = 25$; err.: share of incorrect claims in the stratum; strata with fewer than 20 words omitted).

Dataset	stratum	words	err. (%)	LOCUS	PARSeq agree.	TrOCR agree.
FUNSD	both right	1475	29	99.5	**100.0**	**100.0**
	only TrOCR right	44	80	70.5	47.7	**100.0**
	only PARSeq right	2080	31	98.7	**100.0**	29.0
	neither right	401	93	60.1	56.9	**76.6**
CORD	both right	1592	39	**100.0**	**100.0**	**100.0**
	only TrOCR right	52	71	92.3	61.5	**100.0**
	only PARSeq right	622	35	97.9	**100.0**	34.2
	neither right	90	69	**64.4**	54.4	61.1
SROIE	both right	1946	41	99.5	**100.0**	**100.0**
	only TrOCR right	1827	48	98.1	47.0	**100.0**
	only PARSeq right	33	88	63.6	**100.0**	51.5
	neither right	194	87	73.2	**82.5**	64.4
IAM	both right	1400	13	99.3	**100.0**	**100.0**
	only TrOCR right	296	27	89.5	25.7	**100.0**
	only PARSeq right	1187	57	89.0	**100.0**	43.6
	neither right	1117	82	**81.9**	76.5	73.6
IIIT5K	both right	2764	27	98.9	**100.0**	**100.0**
	only TrOCR right	15	–	–	–	
	only PARSeq right	190	45	86.3	**100.0**	42.6
	neither right	30	97	50.0	53.3	**60.0**

Table S43. Message passing versus counterfactual evidence ($k = 25$, macro-F1 / AUROC, ensembles of five seeds; best in bold, second best underlined). Claim-relative features: the differences $\phi(v) - \phi(h)$ between each rival and the claim and the claim's agreement with the verifiers' readings.

Variant	FUNSD	CORD	SROIE	IAM	IIIT5K
A full	.943 / .974	**.979** / .993	**.973** / **.992**	.901 / **.967**	.970 / **.991**
B no message passing	**.945** / **.975**	.978 / **.994**	.970 / **.992**	.895 / .966	**.979** / **.991**
C no claim-relative features	.936 / .972	.969 / .993	.950 / .988	.898 / .964	.954 / .988
D no claim-relative features, no MP	.943 / .974	.967 / **.994**	.956 / .990	.882 / .960	.960 / .987
E claim node only (no rivals)	.944 / .961	.978 / .993	.972 / **.992**	**.914** / **.967**	.962 / .990

Table S44. Words misread by both verifiers ($k = 25$). err.: share of incorrect claims (%); accuracy (%) of LOCUS, PARSeq agreement and TrOCR agreement; AUROC of LOCUS (cells with fewer than 10 words omitted).

Dataset	stratum	words	err.	LOCUS	PARSeq	TrOCR	AUROC
FUNSD	both wrong, same reading	144	93.8	56.2	51.4	51.4	0.500
	of which claim = that reading	61	100.0	11.5	0.0	0.0	–
	both wrong, different readings	257	92.2	62.3	59.9	90.7	0.638
CORD	both wrong, same reading	47	72.3	57.4	57.4	57.4	0.681
	of which claim = that reading	7	–	–	–	–	–
	both wrong, different readings	43	65.1	72.1	51.2	65.1	0.726
SROIE	both wrong, same reading	27	70.4	51.9	40.7	40.7	0.579
	of which claim = that reading	8	–	–	–	–	–
	both wrong, different readings	167	89.2	76.6	89.2	68.3	0.770
IAM	both wrong, same reading	70	80.0	48.6	44.3	44.3	0.513
	of which claim = that reading	25	100.0	8.0	0.0	0.0	–
	both wrong, different readings	1047	81.9	84.1	78.6	75.5	0.877
IIIT5K	both wrong, same reading	18	100.0	38.9	38.9	38.9	–
	of which claim = that reading	11	100.0	0.0	0.0	0.0	–
	both wrong, different readings	12	91.7	66.7	75.0	91.7	0.727

Table S45. Human-review workload: percentage of all words that must be reviewed so that the automatically accepted words reach the target precision ($k = 25$; lower is better).

Dataset	Method	90%	95%	98%
FUNSD	LOCUS	**30.1**	37.2	61.1
	PARSeq re-OCR (graded)	30.4	**35.4**	**56.2**
	TrOCR-base re-OCR	69.8	100.0	100.0
	Qwen2.5-VL-7B judge	57.1	85.3	100.0
	Engine confidence	47.9	65.5	99.9
CORD	LOCUS	**33.6**	**37.4**	**39.6**
	PARSeq re-OCR (graded)	34.8	38.6	40.6
	TrOCR-base re-OCR	50.8	55.3	57.3
	Qwen2.5-VL-7B judge	36.7	64.9	95.5
	Engine confidence	49.4	59.2	76.0
SROIE	LOCUS	**41.2**	**44.7**	**54.9**
	PARSeq re-OCR (graded)	66.9	69.5	71.7
	TrOCR-base re-OCR	41.9	45.0	94.6
	Qwen2.5-VL-7B judge	53.0	76.2	92.3
	Engine confidence	67.9	90.9	97.0
IAM	LOCUS	**44.3**	**51.4**	66.9
	PARSeq re-OCR (graded)	51.3	54.6	**62.5**
	TrOCR-base re-OCR	99.9	99.9	99.9
	Qwen2.5-VL-7B judge	86.9	94.8	96.8
	Engine confidence	63.9	90.9	99.8
IIIT5K	LOCUS	21.3	25.6	28.1
	PARSeq re-OCR (graded)	**21.1**	**25.3**	**27.7**
	TrOCR-base re-OCR	24.2	28.5	31.0
	Qwen2.5-VL-7B judge	23.0	28.4	34.2
	Engine confidence	26.1	36.9	47.8

S8. Reproduction guide

All steps are scripts in the accompanying code; every table and figure is regenerated from raw result files.

1. **Environment.** Python 3.10, PyTorch 2 with CUDA, PyTorch Geometric, Hugging Face `transformers` (TrOCR, Florence-2, Qwen2.5-VL), PARSeq from its official release, Tesseract 5 and the Liberation fonts.
2. **Data.** Obtain FUNSD, CORD, SROIE, IAM and IIIT5K from their owners; `download.py` and `extract_crops.py` rebuild the crops listed in the released manifests (identical crop ids, splits and seeds).
3. **Claims.** `ocr_hyp.py` runs Tesseract and TrOCR-small and writes the claims and confidences.
4. **Evidence.** `locus2_features.py` builds the counterfactual graphs and the render-normalised evidence under TrOCR-base and PARSeq (one GPU, about 18 words/s); `features.py` computes the character evidence.
5. **Baselines.** `reocr_baselines.py`, `vlm_judge.py` and `eval_external.py`.
6. **LOCUS.** `locus.py main` (pretraining, k-shot adaptation, ensembles), `locus.py ablate`, `xengine` and `xdataset`; the revision analyses are `revision_exp.py` (noise, graph size, pseudo-labels, recalibration), `revision_beam.py`, `revision_bench.py` and `revision_analysis.py`.
7. **Tables and figures.** `make_assets.py`, `make_analysis.py`, `revision_tables.py`, then the LaTeX build script. Each result row in `results/*.jsonl` records dataset, method, k, seed and all metrics.